

HH→4b vs. QCD Classification for Trigger Systems

FastML Hackathon – Challenge 1

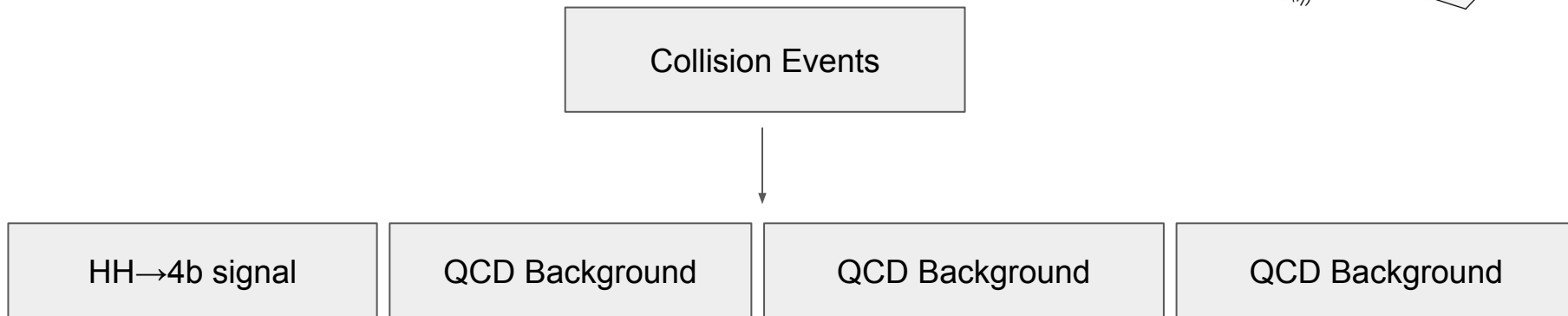
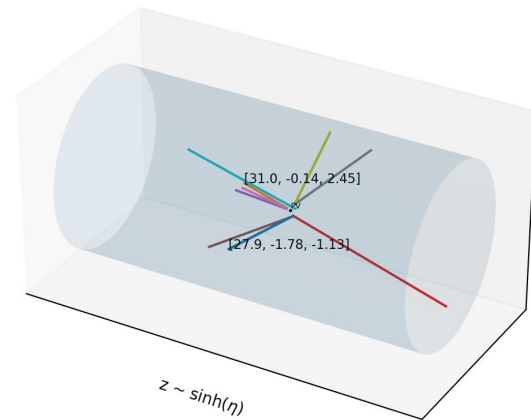
Team name: NID (Nextgen, Important Data analysis)

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NCIPDA (Nextgen Collider, Improved Paradigm-shifting Data Analysis)

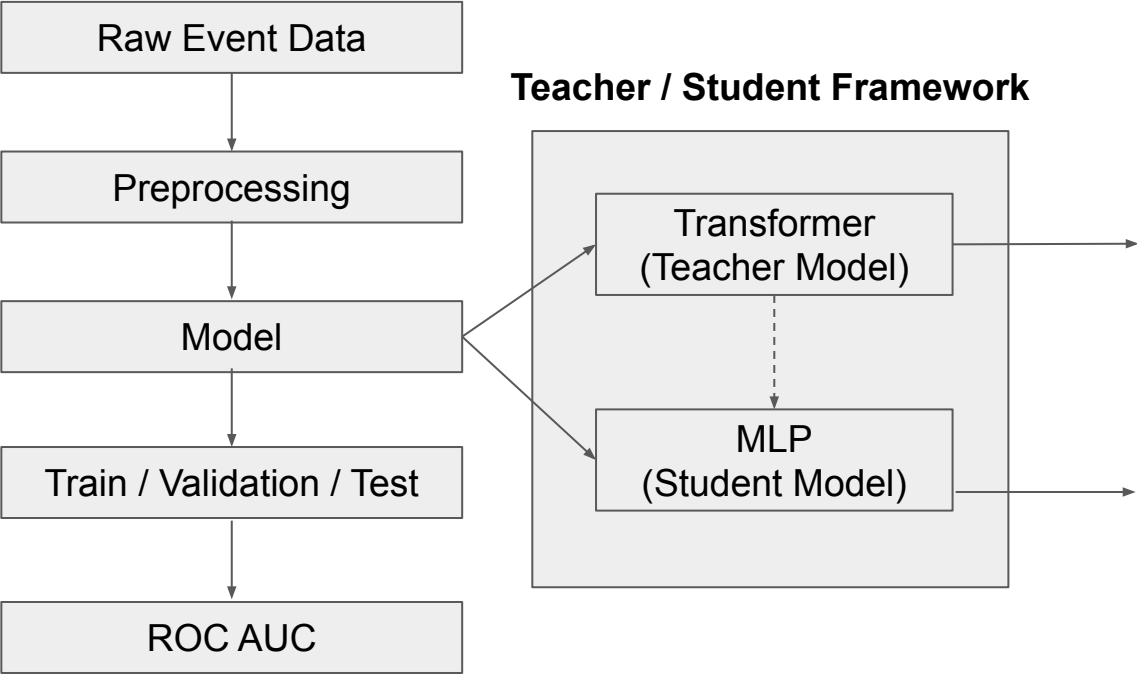
Problem & Motivation

Goal: Identify $HH \rightarrow 4b$ signal events from the dominant QCD background.



Trigger systems must identify interesting events quickly while rejecting the enormous background.

Our Approach



Layer (type)	Output Shape	Param #
input_layer_5 (InputLayer)	(None, 16, 5)	0
dense_12 (Dense)	(None, 16, 64)	384
transformer_block_20 (TransformerBlock)	(None, 16, 64)	33,472
transformer_block_21 (TransformerBlock)	(None, 16, 64)	33,472
transformer_block_22 (TransformerBlock)	(None, 16, 64)	33,472
transformer_block_23 (TransformerBlock)	(None, 16, 64)	33,472
global_average_pooling1d_1 (GlobalAveragePooling1D)	(None, 64)	0
dense_21 (Dense)	(None, 64)	4,160
dropout_34 (Dropout)	(None, 64)	0
dense_22 (Dense)	(None, 32)	2,080
dropout_35 (Dropout)	(None, 32)	0
dense_23 (Dense)	(None, 1)	33

Layer (type)	Output Shape	Param #
input_layer_4 (InputLayer)	(None, 16, 5)	0
flatten (Flatten)	(None, 80)	0
dense_8 (Dense)	(None, 64)	5,184
dropout_16 (Dropout)	(None, 64)	0
dense_9 (Dense)	(None, 64)	4,160
dropout_17 (Dropout)	(None, 64)	0
dense_10 (Dense)	(None, 32)	2,080
dropout_18 (Dropout)	(None, 32)	0
functional_6 (Functional)	(None, 1)	33

Results

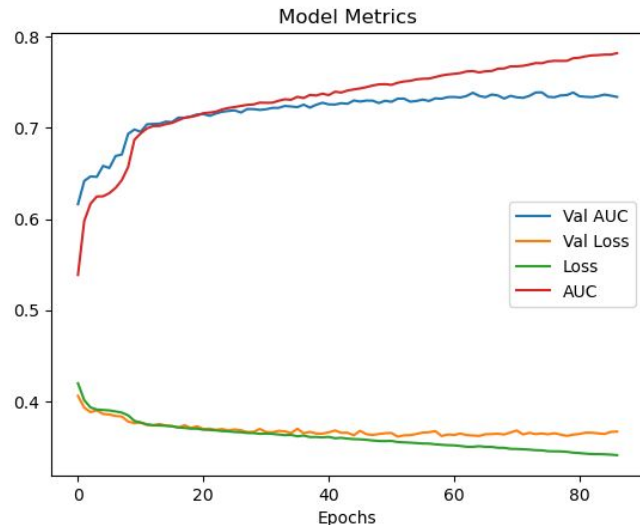
Teacher results:

- AUC: 0.7707
- Loss: 0.3464
- Val AUC: 0.7387
- Val Loss: 0.3648

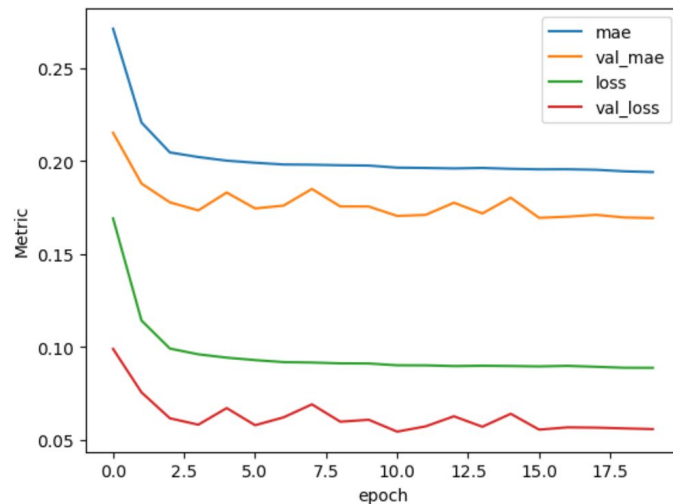
Student results:

- AUC: 0.7143
- Tried post-training the student, but didn't work

Training the teacher



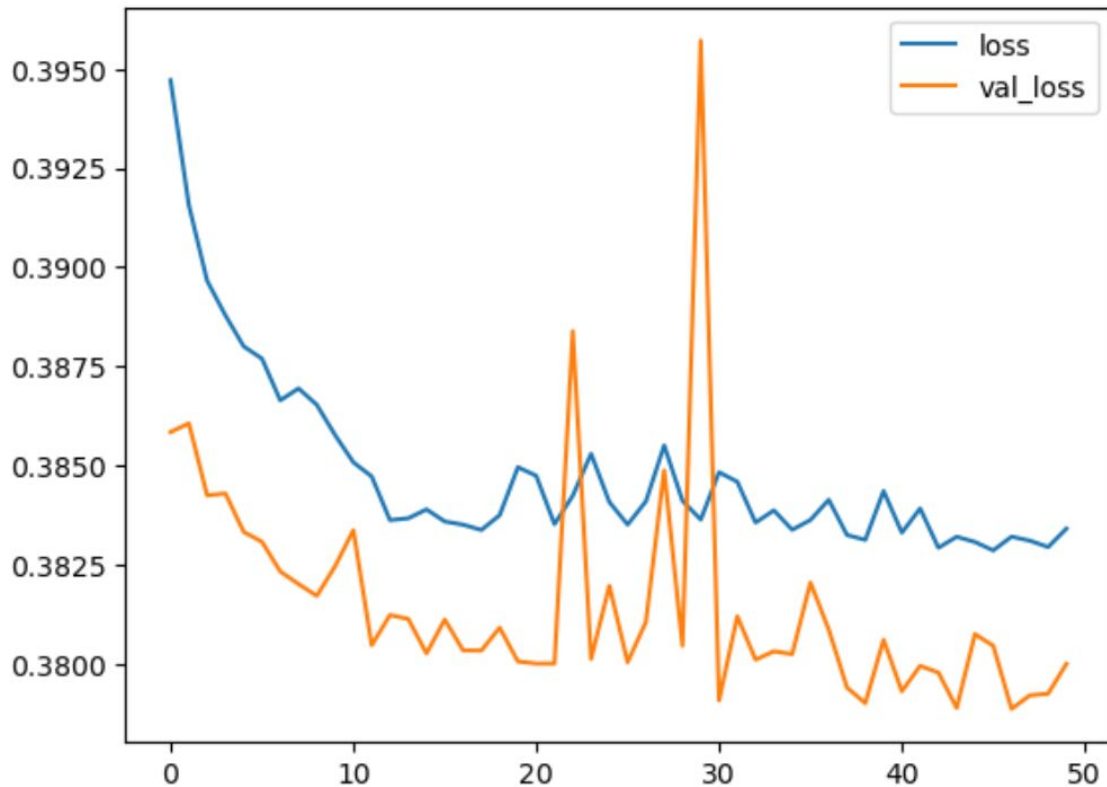
Training the student



Reflection

- Overfitting was consistently a problem and given our limited space capacity on an FPGA we made a very small student model
- Data leakage when using observing more events and increasing generation steps on transformer
- Ran into issues trying to understand dataset as fast as possible, initially trained to identify QCD from jet backgrounds
- Tried HGQ and managed to train a network, but could not get environment to work with hls4ml/keras/jupyterhub and vitis synthesis, so fell back to a Dense student to synthesize, but then didn't have time for that
- With more time, would run a hyperparameter search to explore various teacher models and student models

Backup: post-training the student didn't work



Backup: hgq attempt

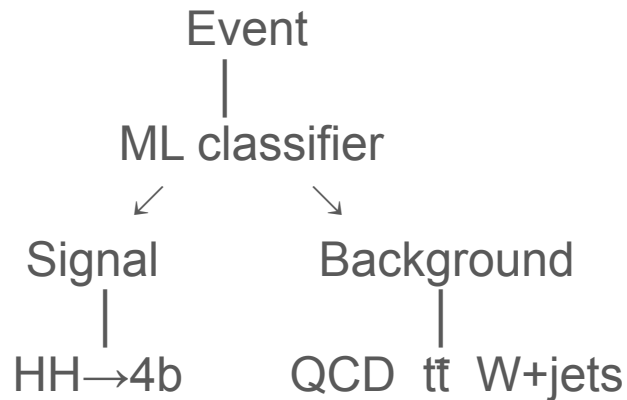
- AUC: 0.6716

Layer (type)	Output Shape	Param #
input_layer_21 (InputLayer)	(None, 16, 5)	0
flatten_1 (Flatten)	(None, 80)	0
q_dense (QDense)	(None, 64)	20,979
dropout_83 (Dropout)	(None, 64)	0
q_dense_1 (QDense)	(None, 32)	8,515
dropout_84 (Dropout)	(None, 32)	0
functional_28 (Functional)	(None, 1)	33

Dataset & Task

Dataset

- 10M $HH \rightarrow 4b$ signal events
- 10M QCD background events
- Additional backgrounds: TT , W +jets
- Trigger-level object features



Student MLP Architecture

